

One customer
One project
One invoice
One project identification code
The system must validate that all four references are consistent.

Payments may be partial, but unidentified or unallocated payments are not permitted in the operational payment workflow.

3.5 Accounts Receivable

Implement accounts-receivable calculations using:

text

Adjusted Amount Due =
Original Invoice Amount
- Approved Credits
- Approved Refunds
- Approved Adjustments
text

Remaining Balance =
Adjusted Amount Due
- Verified Amount Paid

The system must preserve the original invoice amount and each subsequent financial event separately.

3.6 Credits, Refunds, and Adjustments

Implement adjustment records for:

Credits
Refunds
Price adjustments
Other approved adjustments
Each adjustment must include:

Adjustment ID
Project ID
Invoice ID
Adjustment type
Amount
Reason
Requestor
Request date and time
Approval status
Approver
Approval date and time
Effective date
Related payment, when applicable
Related Master Sales Record version, when applicable
Notes

The Sales Manager must approve credits, refunds, and adjustments.

3.7 Accounts-Payable Data Model

Implement payable records with:

Payable ID
Project ID
Vendor ID
Payable category
Description
Amount owed
Certificate ID
Certificate signing date
Payable creation date
Due date
Payment date
Payment method
Amount paid
Outstanding balance
Payment status
Approval status
Created by
Approved by
Approval date
Notes
Related Master Sales Record version
Accounting references
Permitted payable categories:

Equipment
Dealer fee
Materials

3.8 Automatic Payable Creation

When the Certificate reaches the fully signed state, the system must:

Confirm the Certificate status.
Identify the applicable approved costs.
Create separate payables for equipment, dealer fees, and materials.
Use the Certificate signing date as the payable trigger date.
Prevent duplicate payable creation.
Link each payable to the Certificate and project.
Create an audit event.
Display the payable in the accounts-payable workflow.
Payable creation does not require GECC to have paid the vendor.

3.9 Payable Workflow

Implement:

Payable creation
Payable review
Payable approval
Payable payment recording
Partial payable payments

Payable balance calculation

Payable overdue status

Payable dispute status

The Accounts Payable Associate may record payable payments but may not alter the originating project costs, invoice, Certificate, or Master Sales Record.

3.10 Accounting Integration Events

Create ledger-compatible events for:

Verified customer payment

Approved customer refund

Approved credit

Approved invoice adjustment

Payable creation

Vendor payable payment

Payment reversal

The exact chart-of-accounts mapping remains configurable for Sprint 12.

3.11 Reports

Implement:

Accounts-receivable report

Accounts-payable report

Payment-verification report

Adjustment report

Unverified-payment report

Overpayment report

Payables-past-due report

Reports must respect role-based access restrictions.

3.12 Dashboard and Exception Integration

Add dashboard exceptions for:

Unverified cashier's checks

Unverified bank wires

Bank wires missing reference numbers or proof

Rejected payments

Duplicate-payment warnings

Overpayments

Unpaid invoices

Partially paid invoices

Payables pending approval

Payables past due

Payables missing vendors

Refunds awaiting approval

Credits awaiting approval

Adjustments awaiting approval

Projects blocked by missing verified payment

3.13 Audit Integration

Log:

Payment creation

Payment edits

Payment verification
Payment rejection
Payment reversal
Payment refund
Payment allocation
Adjustment creation
Adjustment approval or rejection
Payable creation
Payable edits
Payable approval or rejection
Payable payment
Payable status changes
Financial-record exports
Financial-record downloads
Unauthorized access attempts

4. Approved Sprint Decisions

The following decisions are adopted for Sprint 09 based on the project baseline.

Decision 09-01 — Initial payment methods

The initial implementation supports only:

Bank cashier's checks
Bank wire transfers
Credit cards, ACH, cash, and online payment processing are deferred.

Decision 09-02 — Verification is required

A cashier's check or bank wire is not treated as received cash until verification is complete.

Unverified payments:

Do not reduce accounts receivable.
Do not satisfy the project-payment requirement.
Do not permit project completion.
Do not trigger earned commissions.

Decision 09-03 — Cashier's-check verification

A cashier's check requires bank confirmation before verification.

The confirmation event must identify:

Verifying user
Verification date and time
Bank-confirmation note or reference
Supporting document, when applicable
Decision 09-04 — Bank-wire verification
A bank wire requires either:

A manually entered wire reference number, or
Uploaded proof of receipt
The system must reject verification when neither item exists.

Decision 09-05 — Payment allocation

Every payment must be allocated to a specific project and invoice.

The system must validate consistency among:

- Customer
- Project
- Project identification code
- Invoice

Decision 09-06 — Partial payments
Partial payments are permitted.

The system must calculate remaining balance from verified transactions and approved adjustments rather than from manually edited summary values.

Decision 09-07 — Overpayments
Overpayments must be flagged for resolution.

The system must not automatically treat an overpayment as:

- A new project payment
- A credit
- A refund
- A customer account balance

An authorized user must document the resolution.

Decision 09-08 — Adjustment approval
The Sales Manager must approve credits, refunds, and adjustments.

No adjustment may affect the amount due until the approval event is complete.

Decision 09-09 — Original invoice preservation
The original invoice amount must never be overwritten.

The system must retain:

- Original invoice amount
- Every adjustment
- Adjusted amount due
- Reason
- Approver
- Date and time
- Related records

Decision 09-10 — Payable trigger
Separate payables for equipment, dealer fees, and materials are created when the Certificate is fully signed.

The payable trigger date is the Certificate signing date.

Decision 09-11 — Payable separation
Equipment, dealer fees, and materials must be represented as separate payable records.

They must not be combined into one payable when category-specific source amounts are available.

Decision 09-12 — Payable payment permissions

The Accounts Payable Associate may enter payable payments and update payment status but may not modify the originating project cost or invoice.

Decision 09-13 — Accounting basis

Sprint 09 supports cash-basis reporting by distinguishing:

Invoice issuance

Receivable creation

Payment entry

Payment verification

Cash receipt

Revenue-recognition event

Final chart-of-accounts treatment remains configurable for Sprint 12.

Decision 09-14 — Historical preservation

Payments, refunds, credits, adjustments, payables, and payable payments cannot be silently deleted or overwritten.

Corrections require reversal, replacement, or formal adjustment records.

Decision 09-15 — Audit requirements

All payment and payable actions must use the tamper-evident audit-log system created in Sprint 03.

5. Status Definitions and Transitions

5.1 Payment Statuses

Permitted statuses:

DRAFT

PENDING_VERIFICATION

VERIFIED

PARTIALLY_ALLOCATED

FULLY_ALLOCATED

REJECTED

REVERSED

REFUNDED

CANCELLED

Primary flow:

text

DRAFT

→ PENDING_VERIFICATION

→ VERIFIED

→ PARTIALLY_ALLOCATED

→ FULLY_ALLOCATED

Exception flows:

text

PENDING_VERIFICATION → REJECTED

VERIFIED → REVERSED

VERIFIED → REFUNDED

5.2 Receivable Statuses

Permitted statuses:

NOT_DUE

OPEN

PARTIALLY_PAID

PAID

OVERPAID

DISPUTED

WRITTEN_OFF

`CANCELLED

An invoice may be marked PAID only when verified payments cover the adjusted amount due and no unresolved payment or refund issue remains.

5.3 Adjustment Statuses

Permitted statuses:

DRAFT

PENDING_APPROVAL

APPROVED

`REJECTED

REVERSED

An approved adjustment cannot be silently edited.

5.4 Payable Statuses

Permitted statuses:

DRAFT

PENDING_APPROVAL

`APPROVED

PARTIALLY_PAID

`PAID

`OVERDUE

DISPUTED

CANCELLED

6. Permission Requirements

Sales Associate

May:

View payment status for assigned projects when operationally necessary.

View whether a project is paid, unpaid, partially paid, or awaiting verification.

May not:

Verify payments.

Approve credits, refunds, or adjustments.

View unauthorized financial amounts.

Create or approve payables.

Record payable payments.

Sales Manager

May:

View all payment, receivable, and payable records.
Enter and verify payments.
Approve credits, refunds, and adjustments.
Approve payables.
Review payment exceptions.
Resolve overpayments and payment disputes.
Comptroller
May:

View all payment, receivable, and payable records.
Enter and verify payments.
Review and process financial transactions.
Review payables and payable payments.
Resolve accounting exceptions.
The Sales Manager approval requirement for credits, refunds, and adjustments remains mandatory unless a later approved decision supersedes it.

Accounts Payable Associate
May:

View authorized vendors and payables.
Enter payable payments.
Update payable payment status.
Record payment dates and methods.
View necessary project information.
May not:

Change project pricing.
Change approved project costs.
Approve customer credits or refunds.
Modify invoice terms.
Change the Certificate or Master Sales Record.
Installation Manager
May view installation-related payable information when authorized.

The Installation Manager may not approve customer credits, refunds, or invoice adjustments unless separately authorized by a later approved decision.

Technician and Other Contractors
May not view payment amounts, receivables, payables, refunds, or financial adjustments unless separately authorized.

7. Validation Rules
The system must prevent:

Payment amounts less than or equal to zero.
Unsupported payment methods.
Payments linked to nonexistent projects.
Payments linked to invoices from another project.
Mismatched project identification codes.
Cashier's-check verification without bank confirmation.
Bank-wire verification without a reference number or proof of receipt.
Adjustments without a reason.

Adjustments without Sales Manager approval.

Payables without a project.

Payables without a Certificate reference.

Duplicate payables for the same Certificate, project, category, and source version.

Payable payments greater than the outstanding payable balance unless formally approved.

Financial records being deleted through the application.

Unauthorized access to payment and payable amounts.

Project completion based on unverified payment.

Silent changes to historical financial records.

8. Acceptance Criteria

Sprint 09 is accepted when all of the following are true:

A valid payment can be recorded against a specific project and invoice.

The project identification code is validated against the project and invoice.

Unsupported payment methods are rejected.

Cashier's checks require bank confirmation before verification.

Bank wires require a wire reference number or proof of receipt before verification.

Unverified payments do not reduce accounts receivable.

Unverified payments do not satisfy project-completion requirements.

Partial payments calculate the remaining balance correctly.

Overpayments are detected and flagged.

Original invoice amounts remain unchanged after credits, refunds, or adjustments.

Credits, refunds, and adjustments require Sales Manager approval.

Adjustment reasons and approval history are permanently retained.

Refunds remain linked to the original payment or invoice.

Certificate completion creates separate equipment, dealer-fee, and materials payables.

Payables use the Certificate signing date as the creation trigger date.

Repeated processing of the same Certificate does not create duplicate payables.

Payable balances calculate correctly.

Accounts Payable can record payment date, method, amount, and status.

The Accounts Payable Associate cannot edit originating project costs.

Unauthorized users cannot view restricted financial amounts.

Payment and payable status transitions are enforced.

Payment and payable actions are recorded in the audit log.

Receivable and payable reports are searchable and reproducible.

Unverified payments and overdue payables appear on the exception dashboard.

Verified customer payments create accounting integration events.

Vendor payments create accounting integration events.

Payment reversals and refunds recalculate the affected receivable.

A project cannot become complete using an unverified payment.

Corrections preserve the original financial record.

All automated Sprint 09 tests pass.

9. Acceptance Test Scenarios

Payment Tests

Create a valid cashier's-check payment.

Reject a cashier's-check payment without a check number.

Reject cashier's-check verification without bank confirmation.

Create a valid bank-wire payment.

Reject bank-wire verification without a wire reference or proof document.

Verify a payment and confirm that accounts receivable decreases.

Confirm that an unverified payment leaves accounts receivable unchanged.

Record a partial payment and confirm the remaining balance.

Record an overpayment and confirm the exception state.

Attempt to allocate a payment to an invoice belonging to another project.
Attempt to save a mismatched project identification code.
Attempt to create a duplicate payment and confirm a warning or controlled rejection.
Adjustment Tests
Create a credit request.
Attempt to apply a credit without Sales Manager approval.
Approve a credit and confirm the adjusted amount due.
Create a refund linked to an original payment.
Confirm that the original invoice amount remains unchanged.
Reverse an approved adjustment and confirm the historical adjustment remains visible.
Confirm that all adjustment activity appears in the audit log.

Payable Tests

Fully sign a Certificate and confirm creation of separate payables.
Confirm the equipment payable amount.
Confirm the dealer-fee payable amount.
Confirm the materials payable amount.
Trigger Certificate processing a second time and confirm no duplicates are created.
Approve a payable.
Record a partial payable payment.
Record the final payable payment and confirm PAID status.
Confirm that an overdue payable appears on the exception dashboard.
Confirm that an Accounts Payable Associate cannot edit the source project cost.

Security and Audit Tests

Confirm Sales Associates cannot view unauthorized payment amounts.
Confirm Technicians cannot access payables.
Confirm unauthorized adjustment approval is rejected.
Confirm all access failures are logged.
Confirm all financial changes are present in the tamper-evident audit chain.
Confirm historical records cannot be deleted through the application.

10. Open Questions

The following questions remain open for implementation or later approval. None may weaken the approved baseline.

Open Question 09-01 — Payment entry and verification separation

Must payment entry and payment verification always be performed by different users?

Recommended default: Enforce separation where technically practical, especially for financial control. Permit an authorized exception only with a documented reason and audit entry.

Open Question 09-02 — Overpayment resolution

Should an overpayment be resolved as:

Customer credit

Refund

Invoice adjustment

Manual exception

Recommended default: Require Sales Manager approval and an explicit resolution type. Do not automatically select a resolution.

Open Question 09-03 — Payable due-date calculation

How should payable due dates be calculated?

Possible options:

Fixed number of days after Certificate signing

Vendor-specific payment terms

Manually entered due date

Category-specific terms

Recommended default: Use vendor payment terms when available; otherwise require an authorized user to enter or approve the due date.

Open Question 09-04 — Missing vendor handling

What should happen if the Certificate is signed but a vendor is missing?

Recommended default: Create a controlled VENDOR_PENDING payable exception without permitting payment until the vendor is identified.

Open Question 09-05 — Refund payment method

Should refunds be required to use the original payment method?

Recommended default: Require the refund method to be recorded and require Sales Manager approval when the method differs from the original payment method.

Open Question 09-06 — Payment evidence file types

Which file types should be accepted for proof of receipt?

Recommended default: PDF, PNG, and JPEG, subject to the system's standard file-size and malware-scanning controls.

Open Question 09-07 — Payable overpayment

How should an overpayment to a vendor be handled?

Recommended default: Flag it as an exception and prohibit automatic closure of the payable until corrected or approved.

Open Question 09-08 — Receivable aging intervals

Which aging intervals should be used in reports?

Recommended default: Current, 1–30 days, 31–60 days, 61–90 days, and over 90 days.

Open Question 09-09 — Write-offs

Should write-offs be included in Sprint 09?

Recommended default: Defer write-offs until a formal accounting-policy decision is approved. The data model may reserve a WRITTEN_OFF status without enabling the workflow.

Open Question 09-10 — Accounting event timing

Should the ledger event be created when a payment is verified or when the payment is fully allocated?

Recommended default: Create the cash-receipt event upon verification because the payment is then recognized as received. Allocation validation must still be complete before verification.

11. Sprint 09 Completion Definition

Sprint 09 is complete when:

Payment, verification, receivable, adjustment, payable, and payable-payment workflows operate end to end.

Cashier's checks and bank wires follow their required verification rules.

Payment status controls project completion correctly.

Separate payables are generated from signed Certificates.

Permissions prevent unauthorized financial access and changes.

Adjustments and refunds preserve historical records.

Reports reconcile to underlying transaction records.

Dashboard exceptions are visible to authorized users.

Accounting integration events are generated.

Audit records are append-only and tamper-evident.

Acceptance tests pass.

Open questions are either resolved or formally carried forward.

This bundle is approved and preserved unchanged.

12. Next Markdown Bundle

The next bundle is:

text

GECC-10-costs-and-commissions.md

Its objective is to implement:

Equipment costs

Material costs

Dealer fees

Lead costs

Technician costs

Damon costs

Jerry costs

Gross Profit

Ryan variable commission

Doug variable commission

Net Profit

Sales Associate and GECC profit split

Fixed compensation

Commission eligibility

Biweekly contractor commission reports

Annual contractor summaries

Commission adjustments

Payroll-period corrections

Commission permissions

Commission approval workflow

Integration with completed projects and verified payments

The Sprint 10 bundle must use the verified payment and project-completion statuses established in this Sprint 09 bundle.

===== END FILE: GECC-09-Payments-And-Payables.md =====

=== BEGIN FILE: GECC-10-Costs-And-Commissions.md ===

Bundle name: `GECC-10-costs-and-commissions.md`

Project: GECC Sales Back Office System

Company: Go Ecco Climate Control

Sprint: 10

Sprint name: Costs, Profitability, Commissions, and Contractor Reports

Baseline: GECC-00-project-baseline.md
Previous bundle: `GECC-09-payments-and-payables.md`
Status: Implementation bundle
Date: 2026-09-20
Next planned bundle: GECC-11-inventory-assets-and-loans.md
Primary database: Relational database
System of record: GECC database
Google Sheets: Export or reporting destination only

1. Sprint Objective

Implement project-cost tracking, gross-profit calculations, net-profit calculations, fixed compensation, variable commissions, commission eligibility, commission approvals, biweekly contractor reports, annual contractor summaries, and commission adjustments.

Sprint 10 must ensure that:

Project profitability is calculated from controlled cost records.
Approved costs and corrected costs remain historically traceable.
Commissions are not earned before project completion.
Commissions are not paid on zero-profit or negative-profit projects.
Fixed compensation is calculated according to approved project rules.
Variable commissions use the approved Gross Profit and Net Profit formulas.
Commission reports are generated only for eligible completed projects.
Contractor reports contain only information authorized for the applicable contractor.
Post-payroll corrections are handled through adjustments rather than silent overwrites.

2. Sprint Scope

2.1 Included

Project-cost records
Equipment costs
Material costs
Dealer fees
Lead costs
Technician costs
Damon costs
Jerry costs
Gross Profit
Ryan variable commission
Doug variable commission
Net Profit
Sales Associate and GECC profit split
Fixed compensation
Commission eligibility
Biweekly commission periods
Commission report generation
Commission report approval
Contractor report distribution
Annual commission summaries
Commission corrections
Post-payroll adjustments
Cost and commission permissions
Cost and commission audit logging
Profitability dashboard indicators
Project and customer profitability outputs

2.2 Excluded

Inventory valuation and FIFO implementation
Fixed-asset depreciation
Loan accounting
Full payroll processing
Contractor tax calculations
Bank-statement import
Automatic payment processing
Final financial-statement publication
Vendor payment processing, except integration with Sprint 09
General ledger account configuration, except required integration events
3. Source Records and Dependencies
Sprint 10 depends on:

Approved Master Sales Record versions
Approved retail price
Approved project scope
Equipment records
Material records
Dealer-fee records
Technician cost records
Lead, Damon, and Jerry cost records
Verified customer payments
Approved credits, refunds, and adjustments
Project completion status
Contractor records
User and role permissions
Tamper-evident audit logging
Approved electronic-signature status
Sprint 09 accounts-receivable and payment status
A commission must not be calculated as earned from an unapproved or superseded project version.

4. Approved Cost Categories

The system must support the following initial project-cost categories:

Equipment Cost
Material Cost
Dealer Fee
Lead Cost
Technician Cost
Damon Cost
Jerry Cost
Each cost record must identify:

Cost ID
Project ID
Master Sales Record version
Cost category
Description
Amount
Vendor or contractor, if applicable
Source record
Cost status

Entered by
Entry date and time
Approved by, if approval is required
Approval date and time
Effective date
Revision reference
Correction or adjustment reference
Notes

Amounts must be stored using fixed-precision numeric fields. Floating-point values must not be used for financial calculations.

5. Cost Permissions

5.1 Equipment, Material, and Dealer-Fee Costs

The following roles may enter or revise permitted cost records:

Sales Manager
Installation Manager
Comptroller

5.2 Technician Cost

Technician Cost is a variable actual project cost.

It may be entered or revised by:

Sales Manager
Installation Manager
Comptroller

The system must retain the user, date, reason, and source for each Technician Cost change.

5.3 Lead, Damon, and Jerry Costs

Lead Cost, Damon Cost, and Jerry Cost may be edited for an individual project by authorized users.

The initial default values are:

Lead: \$3,000
Damon: \$500
Jerry: \$250

Changing a cost for one project must not automatically change:

Past projects
Future projects
Other projects
The global default

Any future change to a global default requires separate administrative configuration and must not rewrite historical project costs.

5.4 Sales Associate

A Sales Associate may view limited profitability or commission information only when authorized by the applicable permission policy.

A Sales Associate may view:

Commission information related to their own completed sales

Their own approved commission reports

A Sales Associate may not:

Change approved costs

Change commission calculations

Approve commission changes

Approve commission reports

View another Sales Associate's commission information

5.5 Commission Approvers

The Sales Manager and Comptroller may:

Review commission calculations

Approve commission reports

Review commission corrections

Approve commission adjustments according to the approved workflow

Only the Sales Manager may approve:

Early commission payment

Commission changes

Commission changes caused by corrected project costs

Commission adjustments after payroll approval

6. Profitability Calculations

6.1 Gross Profit

The system must calculate:

text

Gross Profit =

Invoice Amount

- Equipment Cost

- Material Cost

- Dealer Fee

- Lead Cost

- Technician Cost

- Damon Cost

- Jerry Cost

The invoice amount must be the applicable approved invoice amount after authorized pricing changes and before subtracting project costs.

Approved credits, refunds, and adjustments must be handled according to the accounting and profitability treatment approved for the relevant project version. The implementation must not silently substitute a different amount.

6.2 Variable Commissions

Ryan's commission:

text

Ryan Commission = 15%

Doug's commission:

text

Doug Commission = 5% × Gross Profit

The system must store the commission rate used for each calculation so that a future rate change does not alter historical calculations.

If Gross Profit is zero or negative:

No variable commission is paid.

Negative profitability remains visible.

The zero or negative calculation is retained in the commission record.

6.3 Net Profit

The system must calculate:

text

Net Profit =

Gross Profit

-

If Net Profit is zero or negative:

No variable commission is paid.

Negative profitability remains visible.

The negative result is preserved in reports.

6.4 Profit Split

Net Profit is split:

50% to the Sales Associate who created and owns the Master Sales Record

50% to GECC

The system must identify the Sales Associate from the project ownership record, not from a manually entered report value.

If the Sales Associate assignment changes after approval, the change must follow the controlled revision process and must not silently change an already approved commission report.

6.5 Fixed Compensation

Fixed compensation is paid once per completed project to:

Lead

Damon

Jerry

The initial fixed amounts are:

Lead: \$3,000

Damon: \$500

Jerry: \$250

These amounts are project-level deductions and compensation values. They are not dependent on whether Damon or Jerry actively participated in the project, unless a later approved decision supersedes the baseline.

Fixed compensation becomes earned only after project completion.

7. Commission Eligibility

A project is commission-eligible only when all of the following conditions are true:

Required scope of work is complete.

Contract has all required signatures.

Invoice has all required signatures.

Certificate has all required signatures.

Invoice is fully paid after approved adjustments.

Payment verification is complete.

Project status is COMPLETE.

No unresolved payment reversal, refund, or completion-blocking exception exists.

A project that is merely invoiced, signed, installed, or partially paid is not commission-eligible.

7.1 Commission Eligibility Event

When the final eligibility condition is met, the system must:

Create a commission-eligibility record.

Record the eligibility date and time.

Identify the applicable project and Master Sales Record version.

Capture the approved financial inputs.

Calculate fixed compensation.

Calculate variable commissions.

Calculate Net Profit.

Assign the project to the applicable biweekly commission period.

Create a commission-report task.

Record the event in the audit log.

7.2 Eligibility Reversal

If a project later becomes financially incomplete because of:

Payment reversal

Refund

Approved adjustment

Corrected cost

Reopened project

Invalid completion status

the system must not delete the original eligibility record.

Instead, it must create:

Eligibility-reversal record

Reason

Date and time

Approver, when required

Related payment, cost, or adjustment record

Required commission correction task

8. Commission Data Model

8.1 Commission Record

Required fields:

Commission ID

Project ID

Customer ID

Project identification code
Master Sales Record version
Sales Associate ID
Contractor ID
Commission type
Commission rate
Commission basis
Gross Profit
Net Profit
Fixed compensation amount
Variable commission amount
Sales Associate share
GECC share
Eligibility status
Approval status
Payroll status
Report ID
Pay-period ID
Created date and time
Approved date and time
Paid date, if applicable
Correction reference
Adjustment reference
Notes

8.2 Commission Types

Initial permitted values:

FIXED_LEAD
FIXED_DAMON
FIXED_JERRY
VARIABLE_RYAN
VARIABLE_DOUG
SALES_ASSOCIATE_PROFIT_SHARE
GECC_PROFIT_SHARE
COMMISSION_ADJUSTMENT

8.3 Commission Statuses

Permitted statuses:

NOT_ELIGIBLE
ELIGIBLE
`CALCULATED
PENDING_APPROVAL
APPROVED
`REPORTED
PAID
CORRECTION_PENDING
ADJUSTED
VOIDED

A commission record must not be marked PAID unless the applicable commission report has been approved.

9. Commission Calculation Rules

9.1 Calculation Snapshot

When commissions are calculated, the system must store a calculation snapshot containing:

- Invoice amount used
- Approved cost values used
- Gross Profit
- Ryan rate and amount
- Doug rate and amount
- Net Profit
- Fixed compensation amounts
- Sales Associate identity
- Profit-share amounts
- Master Sales Record version
- Calculation date and time
- Calculation engine or rule-set version

Future configuration changes must not recalculate historical reports automatically.

9.2 Rounding

The system must use a consistent financial rounding rule.

Recommended default:

- Calculate using full stored precision.
- Round display and payable amounts to two decimal places.
- Apply final rounding at the individual commission-line level.
- Ensure the report total equals the sum of its displayed line items.
- The selected rule must be documented in the Sprint 10 configuration record.

9.3 Negative Profitability

Negative Gross Profit and Net Profit must be preserved.

The system must:

- Display negative values to authorized users.
- Flag negative-profit projects.
- Suppress variable commissions when Gross Profit is zero or negative.
- Suppress variable commissions when Net Profit is zero or negative.
- Preserve fixed compensation according to the baseline unless a later approved decision changes that rule.

9.4 Missing Costs

A project must not receive a final commission calculation if required cost information is missing.

The system must create a cost-completeness exception identifying:

- Missing cost category
- Project
- Responsible role
- Date detected
- Required action

A Sales Manager, Installation Manager, or Comptroller must resolve the missing cost before final commission approval.

10. Biweekly Commission Reports

10.1 Reporting Period

Contractor commission reports are created every two weeks.

Each report must identify:

Report ID
Contractor
Pay-period start date
Pay-period end date
Report date
Included projects
Included commission records
Total fixed compensation
Total variable compensation
Total adjustments
Net amount reported
Review status
Approval status
Distribution status
Retention reference

10.2 Project Inclusion

A completed project is included in the applicable report when its commission eligibility event falls within the pay period.

A project must not be included solely because:

It was sold during the period.
It was installed during the period.
Its invoice was issued during the period.
Its contract was signed during the period.

10.3 Report Review and Approval

The Sales Manager and Comptroller review commission reports according to the approved workflow.

The implementation must preserve whether:

Sales Manager reviewed
Comptroller reviewed
Sales Manager approved
Comptroller approved
Report was rejected
Rejection reason was provided
The baseline requires Sales Manager and Comptroller review and approval. Both approval events must be retained even if the workflow later permits one authorized final approval for a specific report type.

10.4 Report Distribution

Approved reports must be:

Sent by email to applicable contractors.
Permanently retained.
Linked to the relevant projects.
Linked to the relevant commission records.
Linked to the applicable pay period.

Email delivery status must be retained.

11. Annual Contractor Summaries

Every January 5, the system must create an annual summary for each contractor who was employed during the prior year.

Each summary must include:

Contractor identity

Summary year

Total commission amount

Commission payment dates

Commission amounts

Individual biweekly report references

Distribution status

Delivery date and time

Retention reference

Annual summaries must not expose another contractor's information.

If January 5 falls on a non-business day, the implementation must use the configured delivery policy while retaining January 5 as the reporting trigger date.

12. Commission Adjustments

12.1 Adjustment Triggers

A commission adjustment may be required because of:

Corrected project cost

Payment reversal

Approved refund

Approved invoice adjustment

Incorrect commission calculation

Incorrect contractor assignment

Incorrect project ownership

Commission report correction

Post-approval accounting correction

12.2 Required Adjustment Fields

Adjustment ID

Original commission ID

Project ID

Contractor ID

Original report ID

Original pay period

Adjustment amount

Adjustment type

Reason

Source correction

Requested by

Requested date and time

Approval status

Approved by

Approval date and time

Replacement report or future-period reference

Notes

12.3 Post-Payroll Corrections

If a project cost is corrected after payroll approval:

- Preserve the original commission report.
- Create a commission adjustment.
- Include the adjustment in the next payroll period.
- Add an adjustment note.
- Link the adjustment to the original project.
- Link the adjustment to the original payroll period.
- Require Sales Manager approval.
- Preserve the original calculation snapshot.
- The original approved report must never be silently rewritten.

12.4 Early Commission Payment

Early commission payment requires:

- Sales Manager approval
- Reason
- Project reference
- Amount
- Date
- Audit record
- Early payment must not remove the project from the normal commission-report history.

13. Reporting Requirements

13.1 Project Profitability Report

The report must include:

- Project identification code
- Customer
- Project location
- Sales Associate
- Invoice amount
- Each cost category
- Gross Profit
- Ryan commission
- Doug commission
- Net Profit
- Sales Associate share
- GECC share
- Commission status
- Payment status
- Project completion status

13.2 Customer Profitability Report

The report must aggregate authorized project records by customer and include:

- Customer
- Project count
- Total invoice amount
- Total costs
- Total Gross Profit
- Total Net Profit
- Total commissions

Total adjustments
Total verified payments
13.3 Contractor Commission Report
Each contractor must receive only authorized information.

The report must include:

Contractor identity
Pay period
Eligible projects
Commission type
Commission amount
Approved adjustments
Total amount reported
Approval status
13.4 Negative-Profit Report
The report must identify:

Project
Customer
Sales Associate
Invoice amount
Cost categories
Gross Profit
Net Profit
Variable commission status
Required action
13.5 Commission Adjustment Report
The report must identify:

Original commission report
Original project
Original amount
Corrected amount
Adjustment amount
Reason
Approver
New reporting period
Adjustment status

14. Dashboard Requirements

Sprint 10 must add or populate the following dashboard indicators:

Projects missing required costs
Projects awaiting profitability calculation
Projects with zero Gross Profit
Projects with negative Gross Profit
Projects with zero or negative Net Profit
Projects eligible for commission
Projects awaiting commission approval
Commission reports awaiting review
Commission reports awaiting approval
Commission reports rejected
Commission adjustments pending approval

Post-payroll corrections
Annual summaries awaiting delivery
Failed commission-report emails
Financial amounts and contractor compensation must remain hidden from unauthorized users.

15. Audit Requirements

The system must log:

Cost creation
Cost edits
Cost approvals
Cost reversals
Default-cost changes
Profitability calculations
Commission-calculation events
Commission-rate changes
Eligibility creation
Eligibility reversal
Commission report creation
Commission report review
Commission report approval
Commission report rejection
Commission report distribution
Email delivery failure
Annual-summary creation
Annual-summary distribution
Commission adjustments
Early commission approvals
Unauthorized access attempts
Commission exports, downloads, and prints
Every event must retain the standard audit fields:

User identity
User role
Date and time
Activity type
Related record identifier
Success or failure
Session identifier
Reason when required
Previous audit-log entry reference
Integrity hash or equivalent value

16. Sprint 10 Decisions

Decision 10-01 — Cost categories

The initial implementation uses seven project-cost categories:

Equipment
Materials
Dealer Fee
Lead
Technician
Damon
Jerry

Decision 10-02 — Default fixed deductions

Initial project defaults are:

Lead: \$3,000

Damon: \$500

Jerry: \$250

Defaults are copied into the project cost record when the project reaches the applicable workflow stage. Later default changes do not alter existing projects.

Decision 10-03 — Fixed compensation eligibility

Lead, Damon, and Jerry fixed compensation becomes earned only after project completion.

Decision 10-04 — Variable commission rates

Initial variable commission rates are:

Ryan: 15% of Gross Profit

Doug: 5% of Gross Profit

The rate used must be stored in each commission calculation snapshot.

Decision 10-05 — Zero and negative profitability

No variable commission is paid when:

Gross Profit is zero or negative, or

Net Profit is zero or negative

Negative profitability remains visible in reports.

Decision 10-06 — Net-profit split

Net Profit is split:

50% to the Sales Associate who created and owns the Master Sales Record

50% to GECC

Decision 10-07 — Commission eligibility

Commissions are earned only after:

Scope completion

Full Certificate signature

Full invoice signature

Full payment after approved adjustments

Verified payment

Project completion

Decision 10-08 — Commission report frequency

Contractor commission reports are created every two weeks.

Decision 10-09 — Report retention

Approved commission reports are retained permanently and linked to:

Contractor

Projects

Commission records

Pay period

Distribution events

Decision 10-10 — Annual summary date

Annual contractor summaries are generated every January 5 for contractors employed during the prior year.

Decision 10-11 — Post-payroll corrections

Cost corrections after payroll approval are handled through an adjustment in the next payroll period.

The original report remains unchanged.

Decision 10-12 — Commission approval authority

Only the Sales Manager may approve:

Early commission payment

Commission changes

Cost-correction commission changes

Commission adjustments after payroll approval

Decision 10-13 — Contractor data isolation

Each contractor receives only their own authorized commission information.

Sales Associates see only commission information related to their own sales.

Decision 10-14 — Historical calculation preservation

Historical commission calculations retain their original:

Inputs

Rates

Costs

Profitability values

Ownership

Project version

Calculation rule version

Decision 10-15 — Financial calculation precision

Financial calculations use fixed-precision numeric storage and a single documented rounding policy.

17. Open Questions

Open Question 10-01 — Definition of project invoice amount

Should Gross Profit use:

Original invoice amount, or

Adjusted invoice amount after approved credits, refunds, and adjustments?

Recommended default: Use the approved adjusted economic amount for the applicable completed project version, while retaining the original invoice amount separately. The final accounting treatment must be confirmed before production release.

Open Question 10-02 — Fixed compensation and negative profitability

Should Lead, Damon, and Jerry fixed compensation be paid when a project has zero or negative profitability?

Recommended default: Yes, because the baseline defines fixed compensation separately from variable commissions. This must be explicitly confirmed before payroll implementation.

Open Question 10-03 — Ryan and Doug identities

Are Ryan and Doug fixed contractor identities, or configurable contractor roles that may be reassigned?

Recommended default: Store them as configurable contractor records with a stable commission-role assignment. Historical records must retain the contractor identity used at calculation time.

Open Question 10-04 — Sales Associate reassignment

Who owns commissions when a project is reassigned before completion?

Recommended default: The current approved Master Sales Record owner at the time of commission eligibility receives the Sales Associate allocation. Any reassignment after approval requires a documented revision and approval.

Open Question 10-05 — Commission report approval threshold

Must both the Sales Manager and Comptroller approve every commission report, or may one authorized approver finalize a report?

Recommended default: Require review and approval by both roles because the baseline states that both review and approve commission reports.

Open Question 10-06 — Pay-period boundaries

Should commission periods be based on:

Fixed calendar periods, such as days 1–14 and 15–end of month

Rolling fourteen-day periods

A configurable biweekly calendar

Recommended default: Use configurable fixed fourteen-day periods with a defined first period and permanent period history.

Open Question 10-07 — Rounding method

Should rounding occur:

Per commission line

Per project

Per contractor report

At all three levels with reconciliation

Recommended default: Calculate at full precision, round individual payable lines to two decimals, and reconcile report totals to displayed lines.

Open Question 10-08 — Cost completeness

Which cost categories are mandatory before commission calculation?

Recommended default: All seven baseline cost categories must have a value, including zero where legitimately applicable. Missing values must not be treated as zero without an authorized explanation.

Open Question 10-09 — Commission treatment after refunds

If a refund occurs after commission payment, should the full resulting adjustment be charged to the next contractor report?

Recommended default: Calculate the correction from the affected project and apply it to the next report, subject to Sales Manager approval and applicable legal/accounting review.

Open Question 10-10 — Contractor report delivery failure

What is the required response when an approved report cannot be emailed?

Recommended default: Preserve the approved report, create a delivery exception, notify the Sales Manager and Comptroller, and permit authorized manual delivery.

Open Question 10-11 — Annual report delivery date

If January 5 is a weekend or holiday, should the report be sent:

On January 5

On the preceding business day

On the next business day

Recommended default: Create the annual report on January 5 and deliver on the next configured business day if the delivery service is unavailable.

Open Question 10-12 — Profitability treatment of payment adjustments

Should profitability be recalculated when a payment is refunded or an invoice is adjusted after project completion?

Recommended default: Preserve the original completed-project calculation and create a separate adjustment calculation linked to the original report.

18. Acceptance Criteria

Sprint 10 is accepted when:

All seven approved cost categories can be recorded.

Cost records are linked to the correct project and Master Sales Record version.

Unauthorized users cannot edit approved costs.

Default Lead, Damon, and Jerry costs are applied correctly.

A project-specific cost change does not alter other projects.

Gross Profit is calculated using the approved formula.

Ryan's commission is calculated at 15% of eligible Gross Profit.

Doug's commission is calculated at 5% of eligible Gross Profit.

Variable commissions are zero when Gross Profit is zero or negative.

Variable commissions are zero when Net Profit is zero or negative.

Negative Gross Profit and Net Profit remain visible in authorized reports.

Net Profit is split 50% to the Sales Associate and 50% to GECC.

Fixed compensation is calculated once per completed project.

Fixed compensation is not calculated before project completion.

A project cannot become commission-eligible before all required signatures and verified payment conditions are complete.

A project cannot be included in a commission report solely because it was sold or installed.

Each commission calculation stores its calculation inputs and rates.

Historical calculations do not change when current rates or defaults change.

Missing required costs create a blocking exception.

Biweekly commission reports include only eligible projects in the applicable pay period.

Sales Managers and Comptrollers can review commission reports.

Commission reports require the configured approval workflow.

Approved commission reports are permanently retained.

Reports are linked to the applicable projects, contractors, and commission records.

Contractors receive only their authorized commission information.

Sales Associates see only commission information related to their own sales.

Annual summaries include the required prior-year commission information.

Post-payroll cost corrections create adjustments in the next payroll period.
Original commission reports remain unchanged after corrections.
Only the Sales Manager can approve post-payroll commission adjustments.
Negative-profit projects appear on the appropriate dashboard or exception report.
All cost and commission actions are recorded in the tamper-evident audit log.
Project and customer profitability reports reconcile to underlying records.
Automated Sprint 10 tests pass.

19. Automated Test Scenarios

Cost Tests

Create all seven cost categories for a project.
Confirm unauthorized users cannot edit approved costs.
Change a cost for one project and confirm other projects are unchanged.
Change a default cost and confirm historical projects are unchanged.
Attempt to calculate commission with a missing required cost.
Confirm the missing-cost exception identifies the responsible role.

Profitability Tests

Calculate Gross Profit using positive costs.
Calculate Gross Profit when one cost is zero.
Calculate Gross Profit when total costs exceed the invoice amount.
Confirm negative Gross Profit is retained.
Calculate Ryan's commission at 15%.
Calculate Doug's commission at 5%.
Confirm no variable commission is paid for zero Gross Profit.
Confirm no variable commission is paid for negative Gross Profit.
Confirm Net Profit calculation.
Confirm the 50/50 Sales Associate and GECC split.

Eligibility Tests

Attempt to create a commission before contract signature.
Attempt to create a commission before Certificate signature.
Attempt to create a commission before invoice signature.
Attempt to create a commission before payment verification.
Attempt to create a commission for a partially paid project.
Complete all requirements and confirm commission eligibility.
Reverse a payment after eligibility and confirm an adjustment task is created.

Report Tests

Create a biweekly period.
Include an eligible project in the correct period.
Exclude an ineligible project.
Confirm a project is not duplicated in two reports.
Generate contractor-specific reports.
Confirm one contractor cannot view another contractor's report.
Approve a report and confirm permanent retention.
Reject a report and require a rejection reason.
Generate an annual contractor summary.

Adjustment Tests

Correct a project cost before payroll approval.
Correct a project cost after payroll approval.
Confirm the original commission report remains unchanged.
Confirm an adjustment appears in the next payroll period.
Require Sales Manager approval for the adjustment.
Confirm the adjustment links to the original project and payroll period.

Security and Audit Tests

Confirm Sales Associates cannot view other Sales Associates' commissions.

Confirm Technicians cannot view commission reports.
Confirm unauthorized users cannot export financial data.
Confirm rate changes are audited.
Confirm commission calculation events are audited.
Verify the audit chain after commission corrections.
Confirm financial records cannot be silently deleted.

20. Sprint 10 Completion Definition

Sprint 10 is complete when:

Cost records are controlled and versioned.
Project profitability calculations are reproducible.
Fixed and variable commissions use approved rules.
Commission eligibility is tied to completed projects and verified payment.
Biweekly reports are generated, reviewed, approved, distributed, and retained.
Annual contractor summaries are generated.
Post-payroll corrections use formal adjustments.
Role-based commission visibility is enforced.
Dashboard exceptions are operational.
Audit records are complete and tamper-evident.
Acceptance tests pass.
Open questions are resolved or formally carried forward.
The Sprint 10 bundle is approved and preserved unchanged.

21. Next Markdown Bundle

The next bundle is:

text

GECC-11-inventory-assets-and-loans.md

Its objective is to implement:

FIFO inventory
Serial-number tracking
Equipment acquisition
Equipment allocation to projects
Inventory cost layers
Inventory status transitions
Inventory adjustments
Vehicles
Computers
Offices
Additional fixed-asset categories
Straight-line depreciation
Placed-in-service dates
Accumulated depreciation
Net book value
Asset disposal
Loans
Principal balances
Interest tracking
Loan payments
Inventory, fixed-asset, depreciation, and loan reports
Asset and inventory permissions

Audit integration

Accounting integration with the internal ledger

Sprint 11 must preserve the cost and commission calculation snapshots created in Sprint 10 and must not silently recalculate historical profitability when inventory or asset values are later corrected.

===== END FILE: GECC-10-Costs-And-Commissions.md =====

=== BEGIN FILE: GECC-11-Inventory-Assets-And-Loans.md ===

GECC-11-inventory-assets-and-loans.md

GECC Sales Back Office System — Sprint 11

Bundle Metadata

Bundle name: `GECC-11-inventory-assets-and-loans.md

Project: GECC Sales Back Office System

Company: Go Ecco Climate Control

Sprint: 11

Sprint name: Inventory, Assets, Loans, and Depreciation

Baseline: GECC-00-project-baseline.md

Previous bundle: GECC-10-costs-and-commissions.md

Status: Implementation bundle

Date: 2026-09-20

Next planned bundle: GECC-12-financial-reporting.md

Primary database: Relational database

System of record: GECC database

Google Sheets: Export or reporting destination only

1. Sprint Objective

Implement inventory tracking, serial-number tracking, FIFO cost layers, fixed-asset records, straight-line depreciation, loans, and related accounting integration.

Sprint 11 must ensure that:

Inventory is tracked by product identity and cost layer.

Serialized equipment remains individually identifiable.

Inventory issued to projects is traceable to the project and source cost layer.

Equipment costs used in profitability calculations are reproducible.

Fixed assets are recorded by category and identification details.

Depreciation uses the straight-line method.

Depreciation begins on the placed-in-service date.

Loans track principal, interest, payments, and outstanding balances.

Inventory, asset, depreciation, and loan records are auditable.

Historical project profitability is not silently recalculated after later corrections.

2. Sprint Scope

2.1 Included

Inventory master records

Product identity records

Serialized equipment

Nonserialized materials

Inventory receipts

Inventory cost layers

FIFO issuance

Inventory reservations

Project allocation

Inventory returns

Inventory adjustments

Fixed-asset records

Vehicles
Computers
Offices
Additional asset categories
Asset acquisition
Asset placement in service
Straight-line depreciation
Accumulated depreciation
Net book value
Asset disposal
Loan records
Loan payment records
Principal and interest tracking
Inventory reports
Fixed-asset reports
Depreciation reports
Loan reports
Permissions
Audit logging
Accounting integration events
2.2 Excluded
Automatic vendor purchase-order integration
Automatic bank-statement import
Automatic bank reconciliation
Automatic loan-servicer integration
Accelerated depreciation
Declining-balance depreciation
Section 179 or tax depreciation calculations
Tax-return preparation
Asset maintenance scheduling
Fleet GPS tracking
Technician time tracking
Full procurement management
3. Source Records and Dependencies
Sprint 11 depends on:

Sprint 01 relational database and accounting foundation
Sprint 03 tamper-evident audit logging
Sprint 04 customer, project, and location records
Sprint 05 Master Sales Record and equipment records
Sprint 06 controlled workflow
Sprint 09 payment and payable records
Sprint 10 project-cost and profitability records
Sprint 11 must integrate with:

Equipment records
Project records
Vendor records
Accounts payable
Project-cost records
General-ledger integration events
Financial reporting periods
Audit records

4. Inventory Data Model

4.1 Product Identity

A product identity represents a product type or model.

Required fields:

Product identity ID

Brand

Model

Function

Product type

Description

Unit of measure

Serialized indicator

Active or inactive status

Default vendor, if applicable

Creation date

Last modified date

Examples of product types include:

Condenser coil

Evaporator coil

Refrigerant

Air handler

Furnace

Thermostat

Air ducts

Vents

4.2 Inventory Item

An inventory item represents an individually received or tracked unit.

Required fields:

Inventory item ID

Product identity ID

Serial number, when applicable

Brand

Model

Function

Product type

Acquisition date

Acquisition cost

Vendor

Source payable

Cost-layer ID

Inventory status

Project allocation

Date received

Date issued

Date returned, if applicable

Notes

Created by

Created date and time

Last modified date and time

4.3 Serialized Inventory

Serialized products must retain individual records.

The system must prevent:

Duplicate serial numbers for the same product identity

Issuing the same serial-numbered item to two projects

Changing a serial number without a controlled correction

Deleting a serialized item with historical activity

Reusing a retired or disposed serial number

A serial-number correction must preserve:

Original serial number

Corrected serial number

Reason

User

Date and time

Approval, when required

Related project and document references

4.4 Nonserialized Inventory

Nonserialized materials may be tracked by:

Product identity

Quantity

Unit of measure

Acquisition cost

Cost layer

Vendor

Project allocation

The system must retain transaction-level quantity changes and must not rely solely on a manually edited balance.

5. Inventory Statuses

Permitted inventory statuses:

ORDERED

RECEIVED

AVAILABLE

RESERVED

ISSUED

INSTALLED

` RETURNED

` DAMAGED

SCRAPPED

ADJUSTED

` TRANSFERRED

CANCELLED

5.1 Status Rules

ORDERED items are not available for project allocation.

RECEIVED items have arrived but may require receiving review.

AVAILABLE items may be reserved or issued.

RESERVED items are assigned to a project but have not been issued.

ISSUED items are allocated to a project and included in project cost.
INSTALLED items are confirmed as used in the completed scope.
RETURNED items are removed from project allocation and require condition assessment.
DAMAGED items may not be issued without authorized override.
SCRAPPED items may not be returned to available inventory.
CANCELLED items are retained historically but excluded from available quantities.

6. Inventory Receiving

When inventory is received, the authorized user must record:

Vendor
Receipt date
Product identity
Quantity
Unit cost
Total cost
Serial numbers, when applicable
Related payable or vendor document
Receiving notes
The system must:

Validate the vendor.
Validate product identity.
Validate quantity and cost.
Create or update the applicable inventory records.
Create a FIFO cost layer.
Link the inventory receipt to the payable, when available.
Record the receiving event in the audit log.
Inventory may be received before the related payable is paid.

7. FIFO Cost Layers

7.1 Cost-Layer Requirements

Each inventory receipt creates a cost layer containing:

Cost-layer ID
Product identity ID
Receipt date
Quantity received
Quantity remaining
Unit cost
Total layer cost
Vendor
Source payable
Source receipt
Status

7.2 FIFO Issuance

When inventory is issued, the system must consume the oldest available cost layer first for the applicable product identity.

The system must:

Identify available cost layers.
Order layers by receipt date and cost-layer sequence.
Consume the oldest layer first.

Split the issuance across layers when necessary.
Record the quantity and cost consumed from each layer.
Calculate the total issued cost.
Link the issued cost to the project and Master Sales Record version.
Preserve the cost-layer history.

7.3 Serial-Numbered Equipment

For serial-numbered equipment, the individual serial-number record controls identity and allocation.

FIFO cost-layer tracking must still retain the acquisition cost and receipt sequence, but the system must not substitute one serial-numbered unit for another without a controlled change.

7.4 Cost Corrections

A cost-layer correction must not overwrite the original layer.

The system must create:

Original layer reference
Corrected layer reference
Correction reason
Correcting user
Date and time
Approval status
Affected project references
Accounting adjustment reference

If the affected inventory was already issued to a project, the correction must create a project-cost adjustment rather than silently changing historical project profitability.

8. Project Allocation and Issuance

Inventory may be allocated to a project only when:

The project exists.
The user is authorized.
The product is available or reserved.
The quantity is sufficient.
The project has an approved or otherwise authorized scope requiring the item.
The item is not already allocated to another project.
The allocation record must include:

Allocation ID
Project ID
Master Sales Record version
Product identity
Inventory item or cost layer
Quantity
Unit cost
Total cost
Allocation status
Reserved date
Issued date
Installed date
Returned date, if applicable
User performing each action

Notes

Inventory issued to a project must flow to the project's Material Cost or Equipment Cost according to the item classification.

The system must preserve the distinction between:

Estimated cost

Reserved cost

Issued actual cost

Installed actual cost

Returned cost

Corrected cost

9. Inventory Returns and Adjustments

9.1 Returns

A return must identify:

Project

Product or serial number

Original issue transaction

Return date

Quantity

Condition

Return reason

User

Destination status

Returned inventory may become:

Available

Damaged

Scrapped

Pending inspection

The system must not automatically return an item to available inventory without a condition decision.

9.2 Adjustments

Inventory adjustments require:

Adjustment reason

Quantity or cost difference

Supporting note

Authorized user

Date and time

Approval where required

Related product, serial number, cost layer, or project

Adjustments must be recorded as transactions and must not overwrite the original inventory history.

10. Fixed-Asset Data Model

10.1 Common Fixed-Asset Fields

Each fixed asset must include:

Asset ID

Asset category

Identifying code
Description
Acquisition date
Acquisition cost
Vendor
Source payable
Placed-in-service date
Useful life
Depreciation method
Accumulated depreciation
Net book value
Disposal status
Disposal date
Disposal proceeds
Disposal reason
Location
Responsible user
Created by
Created date and time
Last modified date and time
Asset status
10.2 Vehicle
Required vehicle fields:

VIN
Make
Model
Year
Description
Acquisition date
Acquisition cost
Vendor
Placed-in-service date
Useful life
Accumulated depreciation
Net book value
Location
Disposal information
10.3 Computer
Required computer fields:

Asset serial number
Brand
Model
Description
Acquisition date
Acquisition cost
Vendor
Placed-in-service date
Useful life
Accumulated depreciation
Net book value
Assigned user or location

Disposal information

10.4 Office

Required office fields:

Address

Description

Acquisition date

Acquisition cost

Vendor

Placed-in-service date

Useful life

Accumulated depreciation

Net book value

Disposal information

10.5 Additional Categories

Authorized users may create additional categories containing:

Category name

Identifying code

Description

Acquisition date

Acquisition cost

Placed-in-service date

Useful life

Depreciation data

Creation of a new asset category must be audited.

11. Fixed-Asset Lifecycle

Permitted asset statuses:

PLANNED

ACQUIRED

IN_SERVICE

SUSPENDED

FULLY_DEPRECIATED

DISPOSED

CANCELLED

11.1 Acquisition

An asset is acquired when the asset record is created and linked to the appropriate vendor, payable, invoice, or source document.

11.2 Placed in Service

Depreciation begins on the placed-in-service date.

The system must not calculate depreciation before that date.

The placed-in-service date must be validated as:

On or after acquisition date

A valid calendar date

Recorded by an authorized user

Immutable after depreciation begins except through a formal adjustment

11.3 Depreciation

The only supported depreciation method in this sprint is straight-line depreciation.

The basic calculation is:

text

Depreciable Basis =
Acquisition Cost
- Salvage Value
text

Periodic Depreciation =
Depreciable Basis
÷ Useful Life

The exact periodic convention—monthly, daily, or another approved convention—must be configured before production use.

Accumulated depreciation may not exceed depreciable basis unless a formal correction is approved.

11.4 Net Book Value
text

Net Book Value =
Acquisition Cost
- Accumulated Depreciation
- Approved Impairments or Adjustments

The system must preserve historical depreciation entries and must not overwrite prior-period depreciation.

11.5 Disposal
Disposal requires:

Disposal date
Disposal reason
Disposal status
Proceeds, if any
Authorized user
Approval, if required
Final accumulated depreciation
Final net book value
Disposal accounting event
Audit entry

A disposed asset cannot return to IN_SERVICE without a formal reinstatement process.

12. Loan Data Model
Required loan fields:

Loan ID
Lender

Loan number or reference
Loan amount
Loan date
Term
Interest rate
Payment frequency
Payment schedule
Principal balance
Interest paid
Interest balance, if applicable
Origination fees, if applicable
Maturity date
Loan status
Collateral or related asset, if applicable
Notes
Created by
Creation date and time
12.1 Loan Statuses
Permitted statuses:

PLANNED
ACTIVE
PAID_OFF
DELINQUENT
MODIFIED
DEFAULTED
CANCELLED

12.2 Loan Payments
Each loan payment must include:

Loan payment ID
Loan ID
Payment date
Payment amount
Principal portion
Interest portion
Fees
Payment method
Reference number
User recording payment
Supporting document
Notes

The system must preserve the original payment amount and the allocation between principal, interest, and fees.

12.3 Loan Balance
The system must calculate:

text

Principal Balance =
Original Principal

- Principal Paid
- ± Approved Principal Adjustments

Interest paid must be tracked separately from principal.

A loan may not be marked PAID_OFF until the principal balance and other required amounts are resolved.

13. Permissions

13.1 Sales Manager and Comptroller

May:

- View all inventory, asset, depreciation, and loan records.
- Approve inventory adjustments.
- Approve asset corrections.
- Enter or approve loan records.
- Review accounting effects.
- Resolve exceptions.
- View audit history.

13.2 Installation Manager

May:

- View authorized inventory.
- Reserve or issue installation-related equipment and materials.
- Record project allocation.
- Record actual materials used.
- Propose inventory corrections.
- View installation-related project costs.

May not:

- Modify unrelated fixed assets.
- Change depreciation rules.
- Modify loan records unless separately authorized.

13.3 Accounts Payable Associate

May:

- View authorized inventory receipts and source payables.
- Link inventory receipts to payable records.
- View payable-related asset acquisition information.

May not:

- Change inventory quantities without authorization.
- Change acquisition costs.
- Modify depreciation.
- Modify loans.

13.4 Database Administrator

May:

- Maintain authorized reference data.
- Maintain product identities and asset categories.
- Correct data through controlled administrative procedures.
- View records permitted by the database-administration role.

May not automatically approve financial adjustments unless separately assigned that authority.

13.5 Sales Associates, Technicians, and Other Contractors

May view only inventory or equipment information required for their assigned work.

They may not access:

Unauthorized acquisition costs

Depreciation values

Loan records

Asset financial reports

Inventory valuation reports

14. Accounting Integration

Sprint 11 must create ledger-compatible events for:

Inventory receipt

Inventory issue

Inventory return

Inventory adjustment

Fixed-asset acquisition

Fixed-asset placement in service

Depreciation

Asset disposal

Loan origination

Loan principal payment

Loan interest payment

Loan adjustment

Final chart-of-accounts mapping remains configurable for Sprint 12.

Inventory and fixed-asset corrections must create adjustment events rather than modifying historical ledger events.

15. Required Reports

15.1 Inventory Report

Must include:

Product identity

Brand

Model

Serial number, when applicable

Quantity

Available quantity

Reserved quantity

Issued quantity

Inventory status

Vendor

Cost layer

Unit cost

Total cost

Project allocation

15.2 FIFO Cost-Layer Report

Must include:

Product identity

Cost-layer ID

Receipt date
Original quantity
Quantity consumed
Quantity remaining
Unit cost
Total layer cost
Source vendor
Source payable
Allocated projects
15.3 Fixed-Asset Report
Must include:

Asset ID
Category
Identifying code
Description
Acquisition date
Acquisition cost
Placed-in-service date
Useful life
Accumulated depreciation
Net book value
Status
Location
Disposal information
15.4 Depreciation Report
Must include:

Asset
Period
Beginning net book value
Depreciation expense
Accumulated depreciation
Ending net book value
Depreciation method
Useful life
Placed-in-service date
Adjustment references
15.5 Loan Report
Must include:

Loan
Lender
Original amount
Loan date
Interest rate
Term
Payment schedule
Principal balance
Principal paid
Interest paid
Fees
Maturity date